The Finite-Band Typology of Interface Signatures

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Abstract

The AM-GM Boundary program studies the interface between local viability dynamics and outer aggregation in stochastic systems with extinction. The program's monograph develops the rank-one scalar realization in detail. The present paper, Document 2 of the program, surrounds that realization with a typology of finite-band signatures.

The central technical content is a dissolution. We show that a candidate readout-insensitive finite-band stratum (the program's 2b conjecture) cannot be sustained on rich finite-dimensional comparison domains. The argument has two theorems: ranking-preserving linear readouts on a rich domain are projectively equivalent (readout rigidity); retained-band readouts satisfying the monograph's Axiom A5 must factor the retained dynamics through a one-dimensional quotient, equivalently through a left eigenfunctional of the restricted generator (A5 admissibility). The two theorems close two distinct escape routes the 2b conjecture might have used.

The surviving regions are five: stratum 0 (no stable interface), stratum 1 (rank-one scalar), interpretable stratum 2a (basin-witness scalarization), the formal-cone non-diagnostic region, and stratum 3 (cone-valued comparison without canonical scalar). A dynamic axis $J \in \{\mathrm{qs}, \mathrm{fz}, \mathrm{int}\}$ refines but does not replace the static stratum.

The result is narrower than the program's earlier scalar framing and stronger in formal scope: rank-one scalar compression is one regime among five, and the surviving regions carry substantive partial-order content recorded by signature data. Stratum 3 is the program's forced-pluralism thesis stated mathematically: unanimous-readout comparison on cone-comparable pairs, with rankings of cone-incomparable pairs depending on the declared admissible scalarization class. Four technical arcs — weak-coupling cone and basin stability, matrix-valued Feynman–Kac, dynamic stability of interpretable 2a, and cross-stratum composition — are flagged for subsequent program work. Document 3 will develop empirical signature extraction.

1 Introduction

1.1 What this paper does

The AM-GM monograph [11] establishes the rank-one scalar realization of the AM-GM boundary. Within the killed-process framework, it constructs interface values as positive scalars, proves that geometric outer aggregation is forced from compositional axioms (Theorem II.1 of [11]), exhibits inner arithmetic-totality under complete transfer (Theorem II.2), and verifies the synthesis statically and under slow-fast coarse-state dynamics. Appendix H of [11] identifies the boundary of the scalar regime: when the spectral compression product $(a_2 - a_1)\Delta$ is order one, the principal mode no longer dominates the survival profile, and the natural interface object is a retained finite-dimensional band rather than a positive scalar.

This paper extends that work in a specific direction. *The scalar interface value of Document 1 becomes the typed interface signature of Document 2.* We classify finite-band interface structures by an indexed signature

$$\Sigma(S, P, \Delta, V) = (K, a, \varepsilon_{\text{scalar}}, \varepsilon_{\text{band}}, \mathcal{R}, C, \eta;\ J),$$

indexed by a tuple (S, P, Δ, V) recording system, partition, outer cycle timescale, and viability readout. The resulting assignment partitions supplied signatures into operational regions: stratum 0 (no stable interface), stratum 1 (positive-scalar interface, the monograph's regime), interpretable stratum 2a (vector interface with canonical basin-weighted scalarization), stratum 3 (vector interface with cone-order partial comparison only), and a non-diagnostic formal-cone region between strata 1 and $2\text{a}_{\text{interp}}$ where mathematical positive-cone structure exists without biological interpretive content. The mathematical contribution of the paper is two theorems that close what an earlier version of this program had conjectured was a fifth substantive stratum — a finite-band middle in which multiple scalar readouts agree on rankings — and the typology that survives once that conjecture is closed.

The contribution is narrower than the program's original ambition and stronger in a specific way. Narrower because no single scalar theory of finite-band interface structure exists: generic finite-band systems have provably incomplete comparison structure, and scalarization in such systems is an observer- or measurement-imposed modeling operation rather than a fact supplied by the dynamics. Stronger because the typology is itself the right mathematical frame, not a collection of cases to be unified. The forced-pluralism thesis we defend states this directly: at non-rank-one finite-band timescales, systems with metastable-basin structure may admit canonical scalarization; systems without such structure have vector- or cone-valued interface objects whose scalarization is observer-indexed; under intermediate Feynman–Kac dynamics, this pluralism can amplify rather than reduce.

1.2 The typology, in advance

The signature components are the retained interface coordinate count K (after any admissible recombination of low spectral modes), the spectral data a (decay rates and gaps), the scalarity and band-truncation defects $\varepsilon_{\text{scalar}}$ and $\varepsilon_{\text{band}}$, the admissible readout/scalarization class $\mathcal{R}$, the supplied composition-compatibility data C, the tolerance η carried by the signature itself, and the dynamic regime label $J \in \{\text{qs}, \text{fz}, \text{int}\}$ (quasi-static, fast-coarse-mixing, intermediate; defined in Section 2.3). The signature is downstream of signature extraction: this paper assigns a stratum to a supplied signature; it does not extract signatures from raw biological dynamics. The extraction problem is the empirical methodology of Document 3, named here but not attempted.

The strata themselves are defined in §3. The two central theorems live in §4, where they close two distinct escape routes for a hypothetical readout-insensitive finite-band stratum. The first escape route: if the comparison domain is rich, can multiple linear scalar readouts preserve the same strict rankings without being effectively the same readout? Theorem 4.1 (readout rigidity) shows that ranking-preserving linear readouts on a comparison domain with nonempty relative interior in its affine hull are projectively one-dimensional, hence after normalization a single readout. The second escape route: if such readouts disagree on rankings, perhaps a generic positive scalar readout still satisfies the temporal-sufficiency axiom A5 of the monograph. Theorem 4.3 (A5-admissible readout classification) shows that scalar temporal sufficiency on a rich domain forces the readout, on the relevant invariant span, to vanish on all retained eigenspaces except possibly one eigenvalue cluster, with left-eigenfunctional, degenerate-eigenvalue-cluster, and effectively one-dimensional cases as the exceptional scalar regimes. Together these theorems establish that the stratum the earlier conjecture had named — call it $2b$ in the program's pre-drafting notation — does not exist as a substantive regime. What survives is interpretable stratum 2a (when a basin-witness scalarization is available) and stratum 3 (when it is not).

Stratum 3 carries the program's connection to vector-valued comparison theory. The cone order on a retained band determines a partial order with unanimous-positive-scalarization semantics: cone-comparable pairs are unanimously rankable across all positive scalarizations in the full dual cone $\mathcal{K}^*$ — or across a declared admissible subfamily $\mathcal{R} \subseteq \mathcal{K}^*$, when the signature restricts the readout class — while cone-incomparable pairs are not totally ordered by the dynamics alone, and any scalar ranking among them imports observer or measurement priority. This is forced pluralism stated mathematically: incomplete order structure with substantive content, where the cone records what the dynamics totally order and the admissible class records observer-indexed scalarization within that partial order.

1.3 Adjacent literatures

The mathematical neighbors of this paper differ from those of Document 1, because the central object has changed.

Document 1's neighbors are programs that supply candidate boundaries or compositional aggregation rules at the scalar level. Friston's Free Energy Principle [7, 12] identifies systems by Markov blankets but does not supply an aggregation rule. Krakauer and Flack's information-theoretic individuality [13] measures autonomy but not coherence-as-resource-flow. Garrabrant's geometric rationality [8] identifies the geometric mean as the unique compositional rule across non-transferable branches but does not derive the branches. Peters's ergodicity economics [15, 16] makes the time-average versus ensemble-average distinction precise but does not address subsystem structure. These remain the program-level frame.

Document 2's neighbors are the literatures that handle vector-valued comparison and finite-band spectral structure. Vector optimization [6, 9] supplies the cone-order machinery and the relationship between the dual cone $\mathcal{K}^*$ and admissible scalarization classes. Stochastic ordering [14, 18] supplies parallel partial-comparison structures from a different starting point, defining partial orders by requiring inequalities for all test functions in a class. Eventually-positive semigroup theory [4, 5] supplies language and perturbative tools for formal cone structure on retained spectral bands; its presence motivates the bifurcation between formal and interpretable stratum 2a. Potential-theoretic metastability [2, 3] supplies the existence results for interpretable basin witnesses on which the positive content of stratum 2a rests. Kato perturbation theory [10] and Rudolf–Wang's killed-Markov-process-specific extensions [17] license weak-coupling claims about retained spectral projections, not by themselves about cone or basin-label stability. None of these literatures answers the typology question this paper poses, but each contributes machinery the paper imports or invokes.

The common thread across these neighbors is comparison by families of admissible tests rather than by a single canonical scalar: dual-cone functionals in vector optimization, integral and test-function classes in stochastic ordering, positive cones for semigroups, and basin-local spectral data in metastability. This is the structural reason the typology object is a signature rather than a single number.

1.4 Reading guide

§2 introduces the signature object and the assignment schema, including the precedence convention for tolerance-based stratum assignment. §3 defines the strata, including the explicit distinction between stratum 0 (no stable interface construction at tolerance η) and stratum 3 (stable vector/cone interface, no canonical scalar readout). §4 proves the two central theorems and closes the readout-insensitive-middle conjecture. §5 develops interpretable stratum 2a in detail with the witness criterion, a literature-imported existence proposition, and one negative and one positive worked example. §6 develops stratum 3 with the cone-order proposition and the relation between the full dual $\mathcal{K}^*$ and a declared admissible readout class $\mathcal{R} \subseteq \mathcal{K}^*$. §7 collects operational defects, supplied composition-compatibility data rather than a generic composition law, and type-discipline lemmas. §8 discusses the dynamic regime axis. §9 names technical deferrals — weak-coupling cone stability, dynamic finite-band continuation, dynamic stability of interpretable 2a, mixed-type cross-stratum composition — and program implications for Documents 1 and 3. §10 closes.

Throughout, "classification" means assignment of an already supplied signature to a typological region; extracting such signatures from biological data is the separate Document 3 methodology problem.

2 Signature and typology object

The previous section stated the paper's organizing slogan: the scalar interface value of the AM–GM monograph becomes the typed interface signature of the present paper. This section fixes the notation and type discipline behind that slogan. The basic object is not a raw biological system by itself, but a system viewed under specified indices: a partition, a cycle timescale, and a viability readout. Once those indices have supplied a retained interface structure, the typology assigns that structure to one of the operational regions developed in Section 3.

The important limitation is built in from the start. This paper does not provide an empirical procedure for extracting signatures from raw biological dynamics. It assumes that the relevant spectral, readout, tolerance, and dynamical-regime data have been supplied, estimated, or stipulated by the modeling context. The assignment schema below is therefore a typology of supplied signatures, not a classifier from unprocessed systems.

2.1 The indexed tuple (S, P, Δ, V)

We write

$$(S, P, \Delta, V)$$

for the indexing data relative to which an interface question is posed. Here S names the system under study, P names a candidate partition or coarse-graining of that system into blocks, $\Delta > 0$ is the outer cycle timescale on which blocks are to be evaluated, and V names the viability readout, measurement protocol, or survival-relevant observable being used.

These symbols should be read at the level of modeling inputs rather than as a fully internal formalization of biological dynamics. In a concrete killed-process model, S might include a Markov generator, P a decomposition into block fibers, Δ a cycle length, and V a survival profile or readout functional. In an empirical application, the same tuple records the experimental and modeling choices that determine which interface object is being estimated. The typology is indexed by these choices because the same physical system can occupy different regions at different partitions, timescales, and viability readouts.

This indexing convention does substantive work. It prevents three common slides. First, a typological assignment is not a claim about a system absolutely, but about the system under a specified partition, timescale, and viability readout. Second, a failure of scalarity at one readout does not imply a failure of scalarity at every readout. Third, dynamic regime information is part of the signature rather than an afterthought added once the static classification has been made.

2.2 The signature and its components

For a supplied indexed tuple, the interface signature is written

$$\Sigma(S, P, \Delta, V) = \big(K, a, \varepsilon_{\text{scalar}}, \varepsilon_{\text{band}}, \mathcal{R}, C, \eta; J\big).$$

The semicolon separates static interface data from the dynamic-regime label. The components have the following roles.

Retained interface coordinate count. K is the dimension of the retained interface object after admissible recombination of low spectral modes into interface coordinates. It is not, by convention, the raw count of eigenvalues in whichever theorem is used upstream. In killed or quasi-stationary settings, one often works directly with K basin-local survival profiles. In conservative K-well metastability, the imported low-spectrum theorem may instead describe one equilibrium mode plus $K-1$ transition modes; the K interface coordinates are then the basin profiles recovered after recombination. Section 5 returns to this convention when stating the metastable-basin witness criterion.

Spectral data. a records the decay rates, retained gaps, retained–discarded gaps, and related spectral quantities relevant at cycle length Δ. The exact entries depend on the model class. In the rank-one killed-process regime of the monograph, the key scalarity parameter is the compression product $(a_2 - a_1)\Delta$. In a finite-band regime, the relevant data include the retained cluster and the tail gap controlling band truncation.

Defect terms. $\varepsilon_{\text{scalar}}$ records failure of rank-one scalar compression at tolerance η. Informally, it measures the visible contribution of retained modes beyond the principal scalar mode under the admissible readout class. $\varepsilon_{\text{band}}$ records failure of the retained band to approximate the full interface object. Section 7 gives the operational bounds used for these quantities. At this stage, they are signature data: the typology uses them rather than deriving them from raw dynamics.

Admissible readout class. $\mathcal{R}$ is the declared class of scalar readouts, scalarizations, or measurement functionals admissible for the viability question. The distinction between the full dual cone $\mathcal{K}^*$ and a proper admissible subfamily $\mathcal{R} \subsetneq \mathcal{K}^*$ is part of the signature. Section 6 develops the cone-order semantics; here the point is only that $\mathcal{R}$ is supplied data, not a hidden universal choice.

Composition-compatibility data. C records whatever composition-compatibility data have been supplied. Examples include common basin coordinates, product-basin semantics, a declared scalarization before outer aggregation, or an explicit product-type interface object. The present paper does not prove a generic positive theory of C. Same-type composition under strong shared-coordinate hypotheses appears only as an illustrative construction in Section 7; mixed-type cross-stratum composition is deferred to Section 9. Thus C is a slot for declared compatibility structure, not a promise that the typology itself supplies a canonical cross-block aggregation law.

Tolerance. η is the approximation tolerance with respect to which adequacy claims are evaluated. It travels with the signature. It is not tuned after the fact in order to obtain a desired stratum assignment. A statement such as $\varepsilon_{\text{scalar}} \le \eta$ means: relative to the tolerance already declared in the signature, the scalar approximation is adequate.

Dynamic regime label. J records the dynamic regime. The possible values used in this paper are

$$J \in \{\mathrm{qs}, \mathrm{fz}, \mathrm{int}\},$$

standing for quasi-static, fast-coarse-mixing, and intermediate regimes. The semantics of these labels are introduced below and developed in Section 8. The static strata do not disappear when J is introduced; rather, J records how the static interface type is situated inside a dynamic continuation problem.

2.3 The dynamic regime axis

The dynamic label J is orthogonal to the static stratum assignment. A rank-one scalar interface, an interpretable basin interface, or a generic cone-valued finite-band interface may each appear in different dynamic regimes. The label records how the relevant coarse state behaves over the outer cycle.

In the quasi-static regime, denoted qs, the coarse state is effectively frozen over the cycle. Static spectral and cone data can then be read with minimal dynamic correction. In the fast-coarse-mixing regime, denoted fz, the coarse state mixes rapidly relative to the viability-relevant killing or continuation time, so effective averaged quantities may be available. In the intermediate regime, denoted int, neither reduction is available before the continuation problem is solved. Feynman–Kac or path-dependent structure can then preserve, amplify, or reconfigure finite-band pluralism.

This section does not develop matrix-valued Feynman–Kac theory, adiabatic mode tracking, or dynamic cone transport. Those are future technical arcs. The purpose of J here is only to make the signature honest: dynamic regime data are part of the type of the interface problem, not an optional comment added after the static classification.

2.4 Assignment schema and precedence

We now state the assignment schema used throughout the paper. It is deliberately a schema rather than an empirical classifier. Its input is a supplied signature, including its tolerance η, retained coordinate count K, defect terms, readout class, composition data, and dynamic regime label.

Proposition 2.1 (Typology assignment schema)**.** *Given a supplied signature*

$$\Sigma = (K, a, \varepsilon_{\mathrm{scalar}}, \varepsilon_{\mathrm{band}}, \mathcal{R}, C, \eta; J),$$

assign its static interface type as follows.

1. *If no stable low-dimensional interface construction or viability readout structure exists at tolerance η, assign* stratum 0.

2. *If a positive scalar interface is adequate at tolerance η—in particular if $K = 1$, or if $K \geq 2$ but $\varepsilon_{\mathrm{scalar}} \leq \eta$ under the declared readout class—assign* stratum 1.

3. *If scalar compression is not adequate ($\varepsilon_{\mathrm{scalar}} > \eta$), $\varepsilon_{\mathrm{band}} \leq \eta$, the retained finite-band object carries formal positive-cone structure but no interpretability witness of the kind specified in Criterion 5.1, and no stable vector or cone-valued comparison structure with declared admissible readout class $\mathcal{R}$ is deployed, assign the* formal-cone non-diagnostic region.

4. *If scalar compression is not adequate ($\varepsilon_{\mathrm{scalar}} > \eta$), $\varepsilon_{\mathrm{band}} \leq \eta$, and an interpretability witness exists, assign* interpretable stratum 2a.

5. *If scalar compression is not adequate ($\varepsilon_{\text{scalar}} > \eta$), $\varepsilon_{\text{band}} \leq \eta$, a stable vector or cone-valued retained interface is deployed as comparison structure with admissible readout class $\mathcal{R} \subseteq \mathcal{K}^*$, and no canonical scalarizing witness is supplied, assign* stratum 3*; comparisons are then governed by the cone order on the retained band and the admissible scalarizations in $\mathcal{R}$, rather than by a canonical scalar value.*

The dynamic label J is recorded alongside this static assignment and refines how the assignment is used in continuation problems, but does not replace the static type.

The proposition separates two failures that are easy to conflate. Stratum 0 means that the relevant interface construction itself is not stable at the declared tolerance. Stratum 3 means that a stable vector or cone-valued interface exists, but the dynamics have not supplied a canonical scalar readout. Both lack a canonical scalar interface, but they fail at different levels.

Remark 2.2 (Precedence convention). Assignments use the lowest-dimensional adequate interface type at tolerance η. Thus a finite-band system with $K \geq 2$ and $\varepsilon_{\text{scalar}} \leq \eta$ is operationally stratum 1, while its latent retained-band structure is recorded as secondary signature data. Interpretable stratum 2a and stratum 3 are entered only once scalar compression is not adequate at the declared tolerance.

Remark 2.3 (Witness data deferred). The phrase "an interpretability witness exists" in Proposition 2.1 is unpacked in Criterion 5.1. The witness consists, roughly, of metastable-basin or basin-local data that turn formal finite-band cone structure into biologically interpretable scalarization. Formal cone existence alone is not enough.

2.5 Downstream of extraction

The assignment schema is intentionally downstream of extraction. It says how to type an interface object once the tuple (S, P, Δ, V) has supplied a signature. It does not say how to infer the partition P, the cycle scale Δ, the viability readout V, the retained-band dimension K, the readout class $\mathcal{R}$, or the tolerance η from empirical data. That problem belongs to the empirical methodology of Document 3.

A simple comparison with the monograph makes the point. In the Gibbs-lifted killed-process class of Document 1, when the spectral compression product is large on the outer cycle scale, the supplied signature has an adequate scalar interface and is assigned to stratum 1. At intermediate cycle scales, a one-well potential may have a visible retained band and a formal cone in the physical-state representation, but no basin witness; under the convention above, that places the signature in the formal-cone non-diagnostic region unless a stable vector/cone comparison structure and admissible readout class are explicitly supplied. A multi-well metastable system, by contrast, may supply basin coordinates after recombination and thereby satisfy the witness criterion for interpretable stratum 2a. These examples are developed later; they are mentioned here only to show what the signature is for.

Throughout the paper, therefore, "classification" means assignment of an already supplied signature to a typological region. The typology is a type discipline for interface objects, not a substitute for the biological and statistical work needed to extract those objects.

3 The strata

Section 2 introduced the signature object Σ and stated, as Proposition 2.1, sufficient conditions for assigning a supplied signature to one of five operational regions. This section develops each region:

what interface object it carries, what role it plays in the program, and how it is distinguished from its neighbors. The five regions are stratum 0, stratum 1, the formal-cone non-diagnostic region, interpretable stratum 2a, and stratum 3. They exhaust the typology's static classifications under the precedence convention of Remark 2.2; the dynamic axis J refines but does not replace the static assignment, and its semantics are developed in Section 8.

An earlier version of this program had conjectured a sixth region: a finite-band middle stratum in which multiple scalar readouts agree on rankings without being effectively the same readout. Section 4 proves that this conjecture is false — that no such middle exists. The five regions developed below are the survivors of that proof.

3.1 Stratum 0: when the interface construction itself fails

Stratum 0 collects signatures that fail to admit a stable typological treatment at tolerance η. Concretely, this includes signatures for which no isolated retained band exists on the cycle window, no coherent viability-zero boundary is available, the partition P does not separate the system into stable killed-dynamics blocks, or the admissible readout class $\mathcal{R}$ is not stable across the cycle. In any of these cases, the typology object Σ has been supplied but its components do not jointly support the construction of a finite-dimensional interface object.

The interface object in stratum 0 is undefined. The typology has no positive interface object to analyze for a stratum-0 signature; the assignment is a labeled refusal of interface structure at the declared tolerance. It records that the modeling indices (S, P, Δ, V) have, on this attempt, failed to produce something for the typology to classify. A stratum-0 signature is a tuple under which the typology cannot speak; the underlying system may be perfectly coherent under different indices. A different choice of partition, cycle, or readout may yield a non-trivial classification.

3.2 Stratum 1: scalar interface

Stratum 1 is the regime of the AM-GM monograph. Either $K = 1$ — the retained interface object is one-dimensional from the start — or $K \geq 2$ with scalarity defect $\varepsilon_{\mathrm{scalar}} \leq \eta$ under the declared readout class. In the second case, the precedence convention of Remark 2.2 assigns the signature operationally to stratum 1 and records the latent higher-band structure as secondary signature data.

The interface object is a positive scalar $c \in (0, \infty)$. The weighted geometric mean of Theorem II.1 of the monograph is the canonical outer evaluator across blocks in this regime. The full apparatus of Document 1 — the killed-process realization, the spectral-compression hypothesis on the outer cycle timescale, the Feynman–Kac slow-fast extension — applies to signatures in stratum 1, and the typology delegates compositional questions to that apparatus.

The precedence rule is operationally consequential. A signature with $K = 2$ and tiny but nonzero $\varepsilon_{\mathrm{scalar}}$ is still stratum 1 at sufficiently coarse η. Under a stricter signature with smaller declared η, the same underlying retained-band data may no longer be assigned to stratum 1; whether it lands in interpretable stratum 2a, the formal-cone non-diagnostic region, or stratum 3 depends on whether basin witness data are supplied at the tighter tolerance, and on whether the retained-band structure is being deployed as a comparison object.

3.3 The formal-cone non-diagnostic region

The formal-cone non-diagnostic region collects signatures with $K \geq 2$, $\varepsilon_{\mathrm{band}} \leq \eta$, scalar compression no longer adequate at tolerance η, and a closed eventually-positive cone present on the retained band, but no interpretability witness in the sense of Criterion 5.1 (Section 5) is supplied. The cone exists; its biological meaning does not.

Two features of finite reversible killed dynamics make this region populous rather than exotic. First, the killed semigroup on a finite irreducible state space is positivity-preserving, and at positive times often positivity-improving; a closed positive cone in the physical-state representation typically restricts to an eventually-positive cone on any retained band. Second, the principal eigenfunction of such a generator is strictly positive and gives a canonical Perron readout — mathematically canonical, yet potentially non-diagnostic, because the leading mode of a one-well or otherwise non-multistable system is typically a relaxation mode without basin meaning. The formal cone is an automatic consequence of positivity in the underlying dynamics; the basin meaning is not.

The interface object in this region is a finite-dimensional retained profile equipped with a formal cone, neither operationally scalar in the stratum-1 sense (the scalarity defect exceeds η) nor biologically scalarized in the interpretable-2a sense (no basin-witness data are supplied). Nor is the region operationally stratum 3, because the typology does not deploy cone-order partial comparison without an admissible readout class supplied as comparison structure. The region is the typology's named diagnostic category for formal cones without basin meaning, recorded separately from the four strata. Its role is to prevent an over-strong reading of finite-dimensional positive-cone existence theory: cone existence alone is mathematically real but not biologically diagnostic, and the typology refuses to assign to interpretable stratum 2a on the strength of formal-cone existence alone.

The distinction between formal-cone non-diagnostic and stratum 3 is a typing distinction (signature data) rather than a dynamical one (retained-band properties): the underlying dynamics may be identical, but the typological assignment differs because the modeler has or hasn't declared $\mathcal{R}$ as comparison structure on the retained band.

3.4 Interpretable stratum 2a, previewed

Interpretable stratum 2a is the case in which scalar compression is no longer adequate at tolerance η, but a canonical scalarization is recovered through basin geometry rather than through readout-uniqueness arguments. The signature has $K \geq 2$, $\varepsilon_{\text{band}} \leq \eta$, and an interpretability witness in the sense of Criterion 5.1: metastable basin or basin-module data tied to the viability readout, retained profiles close to basin indicators or basin-local quasi-stationary distributions, a positive cone with non-trivial basin coordinates, and a basin-mass readout compatible with the viability question.

The interface object is a vector in a retained basin cone, equipped with a basin-weighted canonical readout. Composition across blocks with shared basin-coordinate semantics is illustrated by the compatible basin-coordinate construction of Section 7, but is not promoted to a general theorem; cross-stratum aggregation across interpretable-2a blocks and other types is deferred to Section 9 as future work.

The detailed witness conditions are unpacked in Section 5, alongside the literature-imported existence proposition that establishes interpretable 2a as nonempty in a class of metastable reversible killed Markov chains. This section treats interpretable 2a as a label; the witness data behind the label is the work of Section 5.

3.5 Stratum 3

Stratum 3 is the generic finite-band regime in the programmatic sense: the regime expected when neither rank-one scalar compression nor basin-witnessed scalarization is available. The signature has $K \geq 2$, $\varepsilon_{\text{band}} \leq \eta$, a stable vector or cone-valued retained interface, and no canonical scalarizing witness supplied. The cone is present; the dynamics deploy it as the operational comparison structure. Comparisons in stratum 3 follow the cone order on the retained band, with admissible scalarizations in $\mathcal{R} \subseteq \mathcal{K}^*$ as the operational scalarizations. No canonical scalar value collapses this structure.

Concretely, write $\mathcal{K} \subseteq E$ for the closed convex cone on the retained interface space E, with dual cone $\mathcal{K}^* \subseteq E^*$. For $u, v \in E$, the full cone order is $u \succeq_{\mathcal{K}} v$ iff $u - v \in \mathcal{K}$, equivalently $\sigma(u) \geq \sigma(v)$ for all $\sigma \in \mathcal{K}^*$ under the standard closed-convex bipolar hypotheses. If the signature declares a proper admissible class $\mathcal{R} \subsetneq \mathcal{K}^*$, then unanimous ranking over $\mathcal{R}$ defines its own preorder, generally coarser than the cone order in the order-theoretic sense (testing fewer functionals, hence comparing more pairs — the "weaker-test preorder" terminology of Section 6) and not to be identified with the cone order automatically. Cone-comparable pairs are unanimously rankable across the full dual $\mathcal{K}^*$. Cone-incomparable pairs are not totally ordered by the dynamics, and any scalar ranking imposed on them imports observer or measurement priority. Section 6 develops the formal cone-order proposition and the relationship between the full dual $\mathcal{K}^*$ and a declared admissible $\mathcal{R} \subseteq \mathcal{K}^*$.

Stratum 3 is the typology's interface with vector-valued comparison theory. The forced-pluralism thesis previewed in Section 1 states this directly: when the dynamics neither compress the interface to a scalar nor supply a basin witness, comparisons are partial, and the partiality is structural in the dynamics rather than a measurement artifact to be resolved by better instruments.

3.6 Stratum 0 vs stratum 3: failure at different levels

Stratum 0 and stratum 3 share a feature: neither offers a canonical scalar interface. They differ in where, structurally, the canonical scalar fails to exist.

In stratum 0, the interface construction itself is not stable at tolerance η. There is no retained band to compare on, no stable readout class to compare with, no partition that yields stable killed dynamics. The typology cannot deploy any operational comparison rule because there is nothing to deploy it on.

In stratum 3, the interface construction is stable: a retained band exists, a positive cone is present, an admissible readout class is declared, comparisons via cone order are well-defined. What is absent is a basin-witnessed canonical scalarization. The typology operates in stratum 3 with the cone order and admissible scalarizations as the comparison structure; it simply does not collapse this structure into a single scalar value.

The distinction matters because it changes what the typology says about a signature. A stratum 0 signature is typologically un-treated: the partition, cycle, or readout has not yielded an interface object. A stratum 3 signature is typologically classified, and the classification carries operational content — cone-order partial ranking, unanimous comparisons across admissible scalarizations, observer-indexing of cone-incomparable rankings — even in the absence of a canonical scalar.

The next section turns to the proof labor that the typology rests on. The five-region picture above carries proof obligations. The earlier-conjectured sixth region was a substantive proposal, and its dissolution requires proof.

4 Why stratum 2b dissolves

Section 3 described the five surviving regions of the static typology. This section supplies the proof labor behind that list. The dissolution is structural: under the finite-dimensional linear-readout hypotheses appropriate to retained spectral bands, the proposed readout-insensitive middle stratum collapses.

Region	Stable band	Cone	Witness	$\mathcal{R}$ deployed	Canonical scalar
Stratum 0	no	—	—	—	—
Stratum 1	yes	—	—	—	yes (rank-one)
Formal-cone n.d.	yes	yes (formal)	no	no	no
Interpretable 2a	yes	yes (basin)	yes (Crit. 5.1)	—	yes (basin-mass)
Stratum 3	yes	yes (cone/vector)	no	yes ($\mathcal{R} \subseteq \mathcal{K}^*$)	no

Table 1: The five regions of the static typology, indexed by which signature data are stable, witnessed, or deployed. "Formal-cone n.d." abbreviates the formal-cone non-diagnostic region; "—" denotes data not used by the region's classification. Interpretable 2a and stratum 3 are jointly distinguished from formal-cone n.d. by witness data (interpretable 2a) and by deployed comparison structure (stratum 3); the underlying retained-band dynamics may be identical across the three.

The conjectured region was meant to sit between interpretable stratum 2a and stratum 3. It would have been a genuinely finite-band regime: scalar compression would not be adequate, so it would not be stratum 1; but there would nevertheless be a nontrivial family of admissible scalar readouts that all induced the same rankings, so the system would not have the readout-dependence characteristic of stratum 3. In the pre-drafting notation of this program, this was the proposed stratum 2b.

The two theorems below close the two escape routes such a region would need. First, perhaps multiple scalar readouts could agree on rankings without being projectively equivalent. The readout-rigidity theorem rules this out on rich comparison domains. Second, even if scalar readouts disagree on rankings, perhaps a generic positive scalar readout could still be admissible because it satisfies the monograph's temporal interface sufficiency axiom A5. The A5-admissibility theorem rules this out: scalar temporal sufficiency forces an invariant kernel, and hence a one-dimensional factor of the retained dynamics.

4.1 The dissolved conjecture

For this section, a *comparison domain* is a set of retained interface profiles on which the paper asks readouts to rank alternatives. In the intended applications, the ambient space is a retained finite-dimensional interface space, and readouts are linear scalarizations of that retained object. A readout-insensitive finite-band middle would require the following schematic situation.

There is a retained interface space of dimension at least two, scalar compression is not adequate at the declared tolerance, but a nontrivial family of scalar readouts $\mathcal{R}$ preserves all strict rankings on the relevant comparison domain. Thus the finite-band object would not be canonically scalar in the rank-one sense, but it would also not be genuinely pluralistic: all admissible scalar observers would agree on the order of comparisons that matter.

The phrase "that matter" is where the hypotheses enter. If the comparison domain is too small, agreement of readouts is cheap. A one-dimensional comparison domain is already effectively scalar. A domain consisting only of cone-comparable pairs belongs to the partial-order logic of stratum 3. The target of the dissolved conjecture was stronger: a rich finite-dimensional comparison domain with a nontrivial family of readouts that agree anyway. The first theorem shows that this cannot happen for linear readouts.

4.2 The two escape routes

The first escape route is *readout-insensitivity*. It says that a finite-band interface might admit several distinct scalar readouts that nevertheless preserve the same strict rankings. If true on a rich domain, that would support a real middle stratum: scalar readout would not be unique, but ranking would be readout-insensitive.

The second escape route is *temporal sufficiency*. It says that even when generic readouts disagree on rankings, a generic positive readout might still be admissible for outer aggregation because it satisfies the scalar temporal sufficiency principle used in the monograph's outer theorem. This principle is Axiom A5 of Document 1: after temporal subintervals have each been compressed to scalar summaries, whole-interval evaluation must factor through those scalars alone [11]. If a finite-band readout satisfied that condition generically, one might still recover scalar outer composition without rank-one compression.

The two escape routes are independent. The first concerns agreement among readouts on a comparison domain. The second concerns whether a single readout is temporally sufficient for the retained dynamics. Both must be closed for the typology to be stable.

The two theorems also use different richness conditions on the comparison domain D. Theorem 4.1 requires *ranking-richness*: $D - D$ contains a neighborhood of 0 in $H = \operatorname{span}(D - D)$, equivalently D has nonempty relative interior in its affine hull. Theorem 4.3 requires *level-set richness*: $\ker(\ell|_H)$ is spanned by level-set differences $x - y$ with $x, y \in D$ and $\ell(x) = \ell(y)$. These are different geometric requirements; we use the term "rich comparison domain" generically and specify which richness condition is in play at each theorem.

4.3 Readout rigidity

We first prove that rich-domain ranking preservation by linear readouts is projectively trivial. The richness hypothesis is deliberately visible: the comparison domain must have enough local variation that equality of rankings probes the surrounding vector space. Without that hypothesis, the theorem does not apply; the resulting restricted-domain cases are not evidence for a robust readout-insensitive stratum.

Theorem 4.1 (Readout rigidity)**.** *Let V be a finite-dimensional real vector space and let $D \subset V$ be a comparison domain. Set*

$$H := \operatorname{span}(D - D),$$

and assume $\dim H \geq 2$. Assume that D has nonempty relative interior in its affine hull, so that $D - D$ contains a neighborhood of 0 in H. Let $\sigma, \tau \in V^*$ be nonzero linear readouts with $\sigma|_H \neq 0$ and $\tau|_H \neq 0$, such that, for all $x, y \in D$,

$$\sigma(x) > \sigma(y) \iff \tau(x) > \tau(y).$$

Then there exists $c > 0$ such that

$$\tau|_H = c\,\sigma|_H.$$

In particular, on the comparison directions generated by D, the two readouts are projectively equivalent.

Proof. Because $D - D$ contains a neighborhood U of 0 in H, we may test signs of the two readouts on all sufficiently small directions in H. For $u \in U$, choose $x, y \in D$ with $u = x - y$. The hypothesis gives

$$\sigma(u) > 0 \iff \tau(u) > 0.$$

Applying the same implication to $-u$, which also lies in a possibly smaller neighborhood of 0, gives

$$\sigma(u) < 0 \iff \tau(u) < 0.$$

Hence, for every sufficiently small $u \in H$, $\sigma(u) = 0$ if and only if $\tau(u) = 0$.

It follows that the kernels of $\sigma|_H$ and $\tau|_H$ agree. Indeed, if $h \in H$ and $\sigma(h) = 0$, choose $\lambda > 0$ small enough that $\lambda h \in U$. Then $\tau(\lambda h) = 0$, hence $\tau(h) = 0$. The converse is identical. Two nonzero linear functionals on a finite-dimensional vector space with the same kernel are scalar multiples on that space. Thus $\tau|_H = c\sigma|_H$ for some $c \neq 0$. The sign agreement on positive directions in U forces $c > 0$. $\square$

Remark 4.2 (What the richness hypothesis rules out). The theorem is not a claim about every possible comparison set. If H is one-dimensional, the comparison problem has already collapsed to scalarity, which is stratum 1 behavior rather than a finite-band middle. If the domain only asks for comparisons along a cone-comparable subset, then agreement among positive scalarizations reflects the partial-order logic of stratum 3 (formalized in Section 6), not a robust readout-insensitive scalar stratum. The theorem targets the genuine 2b proposal: a rich finite-band comparison domain with multiple non-projectively-equivalent readouts preserving all strict rankings.

The consequence is immediate. A nontrivial family of ranking-preserving linear readouts cannot support an independent finite-band stratum on a rich comparison domain. After normalization, the family has only one projective direction. Either the interface is operationally scalar, or the readout choice matters.

4.4 A5-admissible readouts

Readout rigidity closes the first escape route. The second asks whether a single scalar readout of a finite-band object might nevertheless be admissible because it satisfies temporal interface sufficiency.

In Document 1, Axiom A5 is an axiom on scalar evaluator inputs: once each block has already produced a positive scalar interface value, temporal concatenation must factor through those scalar values. In the retained-band setting the question is prior. Before such an evaluator-level axiom can apply, a finite-dimensional retained interface must produce scalar data through some readout. For $v \in H$ the present state of the retained interface is v, the proposed scalar interface value is $\ell(v)$, and after within-block continuation time t the proposed scalar value is $\ell(T_t v)$. The structural prerequisite for evaluator-A5 in retained-band form is the requirement that this scalar channel be temporally sufficient: future scalar values depend only on the current scalar value, not on the underlying retained state.

The theorem below states the linear-algebraic content of that requirement. A scalar readout is temporally sufficient only if its level sets are invariant under the retained dynamics. Equivalently, the readout factors the retained dynamics through a one-dimensional quotient.

Theorem 4.3 (A5-admissible readout classification)**.** *Let E be a finite-dimensional real vector space and let $T_t = e^{tA}$ be a strongly continuous linear semigroup on E. Let $D \subset E$ be an admissible comparison domain. Let $H \subseteq E$ be a finite-dimensional T_t-invariant subspace containing* $\mathrm{span}(D - D)$ *(equivalently, replace H by the smallest such invariant span). Let $\ell \in E^*$ satisfy $\ell|_H \neq 0$.*

Assume scalar temporal sufficiency on a time set containing an interval with 0 *as an accumulation point: for each such time t, there exists a function ϕ_t such that*

$$\ell(T_t v) = \phi_t(\ell(v)) \qquad (v \in D).$$

Assume also the richness condition that $\ker(\ell|_H)$ *is spanned by level-set differences*

$$x - y, \qquad x, y \in D, \qquad \ell(x) = \ell(y).$$

Then

$$T_t\big(\ker(\ell|_H)\big) \subseteq \ker(\ell|_H) \qquad (t \geq 0).$$

Consequently, $\ell|_H$ *defines a one-dimensional factor of the retained dynamics: on* $H/\ker(\ell|_H)$*, the induced semigroup is scalar multiplication. Equivalently, writing* $A_H := A|_H$ *for the restricted generator,* $\ell|_H \circ A_H = \alpha\, \ell|_H$ *for some* $\alpha \in \mathbb{R}$*; that is,* $\ell|_H$ *is a left eigenfunctional of* A_H*.*

Proof. Let

$$M := \ker(\ell|_H).$$

By the richness hypothesis, it suffices to check invariance on generators of M of the form $x - y$, where $x, y \in D$ and $\ell(x) = \ell(y)$. For such a pair, scalar temporal sufficiency gives

$$\ell(T_t x) = \phi_t(\ell(x)) = \phi_t(\ell(y)) = \ell(T_t y)$$

for every time t in the sufficiency time set. Hence

$$\ell(T_t(x - y)) = 0.$$

By linearity and the spanning assumption, $\ell(T_t h) = 0$ for every $h \in M$ and every such time t. Since H is T_t-invariant, this says $T_t M \subseteq M$ on the sufficiency time set.

Since $T_t = e^{tA}$ on a finite-dimensional space, $t \mapsto \ell(T_t h)$ is real-analytic for each h. Any equality between such functions that holds on a time set with an accumulation point therefore extends to all $t \geq 0$. Because $\ell(T_t h) = 0$ on the sufficiency time set, it vanishes for all $t \geq 0$, and M is invariant under the whole semigroup.

Since H/M is one-dimensional, the induced semigroup on the quotient is multiplication by a scalar $\lambda(t)$. Strong continuity and the semigroup law imply $\lambda(t) = e^{\alpha t}$ for some $\alpha \in \mathbb{R}$. Equivalently, for all $h \in H$,

$$\ell(A_H h) = \alpha \ell(h),$$

so $\ell|_H$ is a left eigenfunctional of A_H. □

Remark 4.4 (Exceptional scalar regimes)*.* When $A|_H$ is diagonalizable, Theorem 4.3 has the expected spectral form. If

$$H = \bigoplus_{\lambda} H_\lambda$$

is the retained eigenspace decomposition, then a left eigenfunctional with eigenvalue α vanishes on every retained eigenspace H_λ with $\lambda \neq \alpha$. Thus an A5-admissible readout is supported, on the relevant invariant span, on at most one retained eigenvalue cluster. The exceptional scalar regimes are therefore left-eigenfunctional cases, degenerate-eigenvalue-cluster cases, and effectively one-dimensional domains. Generic positive mixtures of distinct retained eigenvalue clusters are not temporally sufficient scalar interfaces.

In the reversible/self-adjoint setting emphasized here, $A|_H$ is diagonalizable, so the preceding statement can be read directly in terms of retained eigenspaces. For non-diagonalizable finite-dimensional $A|_H$, the analogous statement is in terms of invariant generalized eigenspaces and the top-of-block functional within each Jordan block at the eigenvalue α; we do not use that generality here.

This theorem is the finite-band version of the monograph's temporal discipline. The monograph's outer characterization applies to scalar interface values only after temporal interface sufficiency has been secured. A retained finite-dimensional profile does not automatically become a scalar input to that theorem merely because a positive readout can be applied to it. Unless the readout collapses the retained dynamics through a one-dimensional factor, it carries unresolved modal memory at the interface.

4.5 Joint verdict

The two escape routes are now closed. The readout-insensitivity route fails because, on rich comparison domains, ranking-preserving linear readouts are projectively equivalent. The temporal-sufficiency route fails because scalar A5-admissibility forces invariant level sets and hence a one-dimensional factor of the retained dynamics. What looked like a possible finite-band scalar middle therefore decomposes into already named cases.

If the finite-band system is operationally scalar at the declared tolerance, it is stratum 1 under the precedence convention of Remark 2.2. If scalar compression is not adequate but the retained modes carry basin-witness data, the system belongs to interpretable stratum 2a; Section 5 develops the witness criterion and the metastable examples that make this region nonempty. If no canonical scalarizing witness is supplied, the system belongs to stratum 3; Section 6 develops the cone-order semantics and admissible-readout discipline for that case. If only formal positive-cone structure exists without interpretability or deployed comparison data, the signature remains in the formal-cone non-diagnostic region described in Section 3.

A residual case deserves explicit attention. A singleton declared readout class $\mathcal{R} = \{\ell\}$ does not by itself restore scalar interface structure. If ℓ is supplied but fails the temporal-sufficiency condition of Theorem 4.3, the resulting scalar ranking is observer- or measurement-indexed comparison data; it is not a canonical scalar interface value. Such a case belongs with the supplied-readout structure of stratum 3 (provided a stable cone-valued retained interface is deployed), not with a resurrected 2b. The singleton $\mathcal{R}$ records observer choice, which the typology classifies as stratum 3 with $|\mathcal{R}| = 1$.

This is the promised dissolution of the proposed stratum 2b. The dissolution is also why finite-band structure must be typed: rank-one scalar interfaces compose by the monograph's geometric evaluator, interpretable basin interfaces require witness-supplied scalarization, and generic finite-band interfaces support cone-order partial comparison. The missing middle is missing because the mathematics of readouts and temporal sufficiency leaves no stable place for it.

5 Interpretable stratum 2a

Section 3 introduced interpretable stratum 2a as the finite-band region in which scalar compression is no longer adequate, but canonical scalarization is recovered from metastable basin structure. This section supplies the promised witness data behind that label. The central point is negative as well as positive: a cone on a retained band is not enough. The retained coordinates must mean something at the viability interface.

The section has three jobs. First, it states the witness criterion used by the assignment schema. Second, it records the standard metastability results that make the criterion nonempty in reversible finite-state settings. Third, it contrasts a one-well non-witness with a multi-well witness. The examples are mathematical models, not biological instantiations; extracting such witnesses from data is the task of Document 3, not of this paper.

5.1 What a witness has to do

The formal-cone non-diagnostic region exists because finite-dimensional positivity is cheap. In finite reversible killed systems, positivity of the physical-state semigroup often induces formal positive cones or eventually-positive structure after projection to a retained band. Such a cone can be perfectly meaningful as linear algebra while still failing to identify biologically interpretable subunits, basins, or persistence modes.

Interpretable stratum 2a asks for more. It asks for a retained finite-dimensional interface object whose coordinates can be read as basin-local or module-local viability profiles, and whose scalar readout is canonical for the viability question because it aggregates those coordinates in a basin-mass or Perron-like way. Both regions have a cone; what distinguishes interpretable 2a is a witness tying the cone to a metastable decomposition relevant to the readout.

There is also a mode-count convention to fix before stating the criterion. The signature coordinate count K is the number of retained interface coordinates after admissible recombination. It is not necessarily the raw eigenvalue count in the external theorem being imported. In a killed or quasi-stationary setting, the low interface data may already appear as K basin-local survival profiles. In conservative K-well metastability, the standard low-spectrum description often consists of one equilibrium mode plus $K - 1$ small transition modes. The K basin coordinates used by the typology are then recovered after recombining the low modes into basin indicators, equilibrium potentials, or basin-local profiles. This convention is how the spectral theorems of metastability talk to the signature Σ.

5.2 The witness criterion

The following criterion specifies the declared data required for a finite-band signature to count as interpretable stratum 2a. A theorem can later show that a model class supplies such data; the criterion itself is a typing rule.

Criterion 5.1 (Interpretable-2a witness)**.** Let a supplied signature have retained interface coordinate count $K \geq 2$, scalarity defect $\varepsilon_{\text{scalar}} > \eta$, and band defect $\varepsilon_{\text{band}} \leq \eta$. An *interpretability witness* for interpretable stratum 2a consists of the following data, all relative to the indexed tuple (S, P, Δ, V) and the declared tolerance η.

(W1) **Basin geometry.** A family of metastable basins, basin modules, or basin-like components

$$\mathcal{B} = \{B_1, \ldots, B_K\}$$

tied to the viability readout V, so that the retained coordinates are meant to track basin-local persistence rather than arbitrary spectral directions.

(W2) **Basin profiles.** A recombination of the retained low-mode data into profiles

$$\psi_1, \ldots, \psi_K$$

on the retained interface space E, with each ψ_α η-close, in the norm appropriate to the model class, to a basin indicator, equilibrium potential, or basin-local quasi-stationary survival profile associated with B_α.

(W3) **Positive basin cone.** A closed convex cone

$$\mathcal{K}_{\mathcal{B}} := \operatorname{cone}\{\psi_1, \ldots, \psi_K\} \subset E$$

with nonempty relative interior in the retained band, so that positive combinations of the ψ_α have the intended interpretation as basin-weighted interface profiles.

(W4) **Canonical basin readout.** A linear functional

$$r_{\mathcal{B}} : E \to \mathbb{R},$$

strictly positive on $\mathcal{K}_{\mathcal{B}} \setminus \{0\}$, compatible with the viability question — for example basin mass, basin-weighted survival, or a Perron/basin-local continuation readout. The readout is canonical relative to the witness data.

(W5) **Cycle-window stability.** The basin family, recombined profiles, cone, and readout remain valid over the declared outer cycle window and tolerance η. Robustness under external perturbations, weak coupling, or cross-stratum composition is recorded only when supplied; otherwise it is future work rather than part of the witness.

When such data are supplied, the retained finite-band object is assigned to interpretable stratum 2a rather than to the formal-cone non-diagnostic region or to stratum 3.

The criterion should be read alongside Section 3. Formal cone existence alone satisfies neither (W1) nor (W2). A generic vector or cone interface without (W4) belongs to stratum 3 once it is deployed as a comparison object with an admissible readout class. Interpretable stratum 2a is the narrower case in which the dynamics and viability question supply the scalarizing witness themselves.

5.3 Metastable existence

We now record the standard source of positive witnesses. The proposition below is an import from the metastability literature, translated into this paper's signature language.

Proposition 5.2 (Metastability supplies interpretable-2a witnesses)**.** *Consider a finite-state reversible Markov chain, possibly killed at a viability boundary, in a metastable regime with K metastable basins $B_1, \dots, B_K$. Assume the standard potential-theoretic metastability hypotheses: transitions within each basin equilibrate rapidly, transitions between basins occur on metastable timescales, the retained low spectrum is separated from the discarded modes by an exponentially large scale separation, and the low modes are asymptotically identified with the metastable states in the sense of Bovier et al. [3] and Bovier and den Hollander [2].*

Then, after the mode-count convention and recombination described in Section 5.1, the retained interface object supplies an interpretability witness in the sense of Criterion 5.1: the basins give (W1), the recombined low modes give basin profiles satisfying (W2), their positive span gives the basin cone (W3), basin mass or basin-local survival gives the canonical readout (W4), and the spectral separation supplies the cycle-window stability (W5) on the metastable window for which the approximation is valid.

Proof sketch / literature import. The cited metastability results identify the low-lying spectrum of reversible metastable chains with transitions among metastable states. In the conservative setting, the low cluster consists of the equilibrium mode together with $K-1$ slow transition modes; after recombination, these modes yield functions close to indicators or equilibrium potentials of the metastable basins. In killed or quasi-stationary settings, the corresponding retained profiles are basin-local survival or quasi-stationary profiles. The separation between the low cluster and the discarded spectrum controls the error of the retained-band approximation over the metastable window.

Thus the imported theorems supply exactly the data required by the witness criterion: a basin family, basin-local profiles, a positive cone generated by those profiles, and a natural basin-mass or basin-local survival readout. The proposition adds only the typological interpretation of those data. □

The proposition is intentionally scoped to reversible metastability. Non-reversible metastable extensions, weak-coupling stability of basin cones, and dynamic transport of witnesses under moving coarse states are not used here. They are future-work arcs named in Section 9.

5.4 A negative example: one well, formal cone, no witness

Example 5.3 (One-well finite band: formal-cone non-diagnostic)**.** Take the one-well Gibbs-lifted killed-process class from Document 1: a reversible finite birth–death chain on a fixed-total fiber, with a single convex potential well centered away from the killing boundary and threshold killing near the endpoints [11]. On long enough cycle windows the principal killed mode dominates. By the precedence convention, the signature is then stratum 1.

Now shorten the cycle window so that the second mode remains visible while the discarded modes are still controlled. The retained object is no longer adequately scalar, and one can form formal positive-cone structure in the physical-state representation. But the second visible mode is an intrawell relaxation direction: in the symmetric case it is the familiar signed left-versus-right relaxation mode inside one basin. It is not close to an indicator of a second metastable basin, nor to a basin-local quasi-stationary survival profile. There is no family $\{B_1, B_2\}$ satisfying (W1), no recombination into two basin-local profiles satisfying (W2), and no basin-mass readout satisfying (W4).

Thus the intermediate one-well signature is not interpretable stratum 2a. By default it belongs to the formal-cone non-diagnostic region. It becomes a stratum 3 comparison object only if the modeler explicitly deploys the retained cone as comparison structure and supplies an admissible readout class $\mathcal{R}$. Formal Perron structure alone does not promote the example to interpretable 2a.

This example is the reason the witness criterion is needed. Without (W1)–(W4), an eventually-positive finite-band cone could be mistaken for biological or viability-relevant scalarization. The typology refuses that move: formal positivity and interpretable basin scalarization are different data.

5.5 A positive example: a finite-state multi-well witness

Example 5.4 (Finite-state multi-well chain: witness supplied)**.** Let Ω be a finite state space decomposed into K disjoint basins

$$\Omega = B_1 \sqcup \cdots \sqcup B_K.$$

Consider a reversible continuous-time Markov chain whose generator has the form

$$L_\varepsilon = L_{\mathrm{fast}} + \varepsilon L_{\mathrm{slow}}, \qquad 0 < \varepsilon \ll 1.$$

Here L_{fast} mixes rapidly inside each basin and does not move mass between basins, while L_{slow} gives rare reversible transitions among basins. Add, if desired, killing at an external viability boundary with killing rates small on the metastable window. This is a schematic finite-state version of the reversible metastable situation treated by Bovier et al. [3] and Bovier and den Hollander [2].

The witness data are explicit. The basin family $\mathcal{B} = \{B_1, \ldots, B_K\}$ gives (W1). For small ε, the retained low modes are recombined into profiles ψ_α close to the basin indicators 1_{B_α}, or in the killed version to basin-local quasi-stationary survival profiles. This gives (W2). The cone

$$\mathcal{K}_{\mathcal{B}} = \left\{ \sum_{\alpha=1}^{K} c_\alpha \psi_\alpha : c_\alpha \geq 0 \right\}$$

with coordinates c_α gives (W3). A basin-mass readout, for example

$$r_{\mathcal{B}}\left(\sum_{\alpha=1}^{K} c_\alpha \psi_\alpha\right) = \sum_{\alpha=1}^{K} m_\alpha c_\alpha, \qquad m_\alpha > 0,$$

where m_α records the viability-relevant mass, prevalence, or survival weight of basin B_α, gives (W4). The retained–discarded spectral gap and the separation between intrabasin equilibration and interbasin motion give (W5) on the declared cycle window.

This example is interpretable stratum 2a at tolerances for which scalar compression is not adequate but the retained basin band is accurate. The scalarization is canonical because the witness data tie the retained coordinates to metastable basins and to the viability readout.

The positive example also explains why interpretable 2a is narrow. If the ψ_α cease to be basin-local, if the retained–discarded gap closes, if the readout is changed so that basin mass is no longer viability-relevant, or if cross-block composition requires coordinates that are not shared, the witness no longer supplies what the assignment schema needs. The system may fall to the formal-cone non-diagnostic region or to stratum 3, depending on whether a stable cone comparison structure is actually deployed.

5.6 What interpretable 2a promises

Criterion 5.1 and Proposition 5.2 establish the nonemptiness of interpretable stratum 2a in a class of mathematical systems. They do not claim that any particular biological system instantiates the criterion. In particular, a biological example requires evidence for the indexed tuple (S, P, Δ, V): the partition, cycle window, viability readout, retained-band dimension, basin profiles, admissible readout, and tolerance all have to be extracted or justified. That is Document 3 territory.

Nor does interpretable 2a provide a generic cross-block composition theorem. It supplies a canonical scalarization within a witnessed basin-coordinate interface. Composition across multiple witnessed blocks requires supplied compatibility data C, such as shared basin-coordinate semantics or an explicit product-basin interpretation. Section 7 gives only an illustrative construction under such supplied data; mixed-type cross-stratum composition is deferred to Section 9.

The payoff is therefore precise. Interpretable stratum 2a is the finite-band case where scalarization is earned by metastable basin semantics. When those semantics are absent, the typology does not force a scalar; it sends the signature either to the formal-cone non-diagnostic region or, if a cone comparison structure is supplied, to stratum 3.

6 Stratum 3 and forced pluralism

Section 3 described stratum 3 as the regime of stable vector or cone-valued retained interfaces with no canonical scalarizing witness. This section develops the partial-order content of that regime: the cone-order proposition that organizes comparisons in stratum 3, the relationship between the full dual cone $\mathcal{K}^*$ and a declared admissible readout class $\mathcal{R}$, and the mathematical content of the program's forced-pluralism thesis. The connections to vector optimization and stochastic ordering are made explicit because they are where the typology's partial-order machinery lives in the broader literature.

6.1 Setup: vector interface, no canonical scalar

A signature in stratum 3 has retained interface coordinate count $K \geq 2$, finite-band defect $\varepsilon_{\text{band}} \leq \eta$, scalar compression not adequate at tolerance η, and a stable vector or cone-valued retained interface deployed as comparison structure with declared admissible readout class $\mathcal{R}$. The retained interface space is E, dimension K. The closed convex cone $\mathcal{K} \subseteq E$ is supplied by the dynamics, typically as an eventually-positive cone arising from the killed semigroup's restriction to the retained band, or as a cone of positive coordinates in a basin-coordinate representation when basin geometry exists but is non-witnessing. The admissible readout class $\mathcal{R}$ is signature data: it records which nonzero positive scalarizations of the retained band, considered up to normalization, are admissible for the viability question.

The comparison question for stratum 3 is: given $u, v \in E$, what ranking can the typology assert? The answer depends on the supplied readout class. Unanimous-$\mathcal{R}$ ranking, defined by $\sigma(u) \geq \sigma(v)$ for all $\sigma \in \mathcal{R}$, gives a preorder on E. When $\mathcal{R} = \mathcal{K}^*$ this preorder coincides with the cone order on E developed below. When $\mathcal{R} \subsetneq \mathcal{K}^*$, unanimity over $\mathcal{R}$ tests fewer functionals than unanimity over $\mathcal{K}^*$, and pairs satisfying $\mathcal{R}$-unanimity need not be cone-comparable in $\mathcal{K}$. Ranking via a single $\sigma \in \mathcal{R}$ gives a total preorder by scalar score, but it is observer-indexed: the supplied $\mathcal{R}$ singled out σ, the dynamics did not.

6.2 The cone-order proposition

The cone-order proposition organizes ranking under the full dual cone. It is a finite-dimensional convex-analysis result imported from the bipolar theorem; the proposition is stated here in the typology's notation for completeness.

Proposition 6.1 (Cone-order proposition)**.** *Let E be a finite-dimensional real vector space, and let $\mathcal{K} \subseteq E$ be a closed convex cone with dual cone*

$$\mathcal{K}^* := \{\sigma \in E^* : \sigma(k) \geq 0 \textit{ for all } k \in \mathcal{K}\}.$$

Then for $u, v \in E$,

$$u - v \in \mathcal{K} \iff \sigma(u) \geq \sigma(v) \textit{ for all } \sigma \in \mathcal{K}^*.$$

Proof. By the bipolar theorem for closed convex cones in finite-dimensional spaces [1], $\mathcal{K} = \mathcal{K}^{**}$, where

$$\mathcal{K}^{**} = \{w \in E : \sigma(w) \geq 0 \text{ for all } \sigma \in \mathcal{K}^*\}.$$

Applied to $w = u - v$, this gives

$$u - v \in \mathcal{K} \iff u - v \in \mathcal{K}^{**} \iff \sigma(u - v) \geq 0 \text{ for all } \sigma \in \mathcal{K}^* \iff \sigma(u) \geq \sigma(v) \text{ for all } \sigma \in \mathcal{K}^*.$$

□

Remark 6.2 (Antisymmetry under pointedness)*.* The relation $u \succeq_{\mathcal{K}} v$ defined by $u - v \in \mathcal{K}$ is reflexive and transitive in general. It is antisymmetric — and so a partial order on E — when $\mathcal{K}$ is pointed, that is, $\mathcal{K} \cap (-\mathcal{K}) = \{0\}$. In the model classes emphasized here, supplied cones are intended to be pointed; non-pointed supplied cones are recorded as signature data, with the induced relation a preorder lacking antisymmetry.

6.3 Full dual versus declared admissible class

The signature's admissible readout class $\mathcal{R}$ records which positive scalarizations are admissible for the viability question. Its relationship to the full dual cone $\mathcal{K}^*$ is consequential for what the typology asserts.

Remark 6.3 (Full dual versus declared admissible class). For a signature in stratum 3, the cone $\mathcal{K} \subseteq E$ is supplied by the dynamics, and the admissible readout class $\mathcal{R} \subseteq \mathcal{K}^*$ is supplied by the signature. Two cases are operationally distinct.

When $\mathcal{R} = \mathcal{K}^*$, unanimous-$\mathcal{R}$ ranking equals the cone order from Proposition 6.1. Comparisons that are unanimous across $\mathcal{R}$ are exactly the cone-comparable comparisons.

When $\mathcal{R} \subsetneq \mathcal{K}^*$, unanimous-$\mathcal{R}$ ranking defines a generally different, weaker-test preorder on E: because fewer scalarizations are tested, it may rank pairs that are not comparable in the full cone order. The $\mathcal{R}$-induced preorder records which comparisons are unanimous over the supplied admissible class, and is genuinely distinct from the cone order.

The supplied $\mathcal{R}$ thus records how restricted the observer's scalarization choices are; the cone $\mathcal{K}$ records how restricted the dynamics' positivity structure is. These are independent parameters of a stratum-3 signature, and the typology keeps them separate.

6.4 Connections to vector optimization and stochastic ordering

Stratum 3 is the typology's interface with two adjacent literatures.

Vector optimization. Vector optimization [6, 9] studies decision problems in which alternatives are compared by multiple criteria simultaneously. Pareto-optimal solutions are alternatives in a feasible set that no other feasible alternative dominates in the chosen cone-order sense. Scalarization theory recovers total orderings by constructing scalar-valued objectives that respect the cone order, parameterized by weights or admissible scalarizations. The typology's stratum 3 inherits this structure: the supplied admissible readout class $\mathcal{R}$ is the set of scalarizations admissible for the viability question. What is distinctive in the typology's framing is the indexing: the cone $\mathcal{K}$, the admissible class $\mathcal{R}$, and the comparison-domain question are all signature data, indexed by the modeling tuple (S, P, Δ, V).

Stochastic ordering. Stochastic-orders theory [14, 18] defines partial orders on probability distributions by requiring inequalities to hold for all test functions in a declared class — for example, all increasing functions (the usual stochastic order $\leq_{\mathrm{st}}$), all increasing convex functions (the increasing convex order $\leq_{\mathrm{icx}}$), or more general integral-stochastic orders. The structural analogue in stratum 3 is direct: unanimity over a declared class of admissible scalarizations defines a partial order on retained interface profiles. The typology's $\mathcal{R}$ plays the role of the test class. The interpretation differs: stochastic-orders test classes typically encode utility-theoretic or decision-theoretic preferences, while $\mathcal{R}$ in the typology encodes which observers' viability scalarizations are admissible.

These literatures supply machinery the typology imports. Vector optimization gives the cone-order plus scalarization treatment; stochastic ordering gives the unanimity-over-test-classes treatment. The typology's contribution is the typing discipline: which signatures should be classified to stratum 3, what the supplied $\mathcal{R}$ records about observer-indexing of scalar rankings, and how stratum 3 sits in the larger five-region static partition.

6.5 The forced-pluralism thesis

The cone-order content of stratum 3 has substantive interpretation. Forced pluralism is the program's name for the mathematical content developed above.

When the dynamics neither compress the retained interface to a scalar (stratum 1) nor supply a canonical basin-witness scalarization (interpretable stratum 2a), comparisons are governed by the cone order together with the admissible readout class. Cone-comparable pairs u, v with $u - v \in \mathcal{K}$

are unanimously rankable across all admissible scalarizations $\sigma \in \mathcal{R} \subseteq \mathcal{K}^*$. Cone-incomparable pairs are not totally ordered by the full cone order; if a restricted $\mathcal{R}$ ranks them, that ranking is supplied by the admissible readout choice in $\mathcal{R}$, recording observer or measurement priority that is itself part of the signature data.

The pluralism is forced in a precise sense: the partiality of the order is structural in the dynamics, recorded by the cone $\mathcal{K}$, with the supplied $\mathcal{R}$ recording how much further observer choice has narrowed the admissible scalarizations. Tightening measurement or refining the readout class may move a signature out of stratum 3 — into interpretable 2a if a basin witness becomes available, or into stratum 1 if a tightened tolerance reveals scalarity. Within a fixed signature, however, cone-incomparability is structural: it records the limits of what the dynamics totally order, and tightening measurement does not eliminate it.

The forced-pluralism thesis as developed here is mathematical content. The typology's content is the cone, the admissible class, the partial order they jointly determine, and the typing discipline that classifies signatures by these data. Philosophical bridges to broader accounts of patternhood or biological individuality are program-level work elsewhere.

7 Operational defects and supplied composition data

The preceding sections define the typology's regions and prove why the readout-insensitive finite-band middle does not survive. This section collects operational machinery used across the typology: how the defect terms should be read, how scalarity defect is bounded uniformly over an admissible readout class, what the composition-compatibility slot C records, and what continuity of retained spectral subspaces does and does not buy. None of these ingredients is a new stratum. They are the bookkeeping and type discipline that make the signature usable.

7.1 Operational defects: the prose-only stack

The signature contains several quantities that measure different ways an interface description can fail to be exact. They should not be collapsed into a single naturality theorem. The typology uses them as a stack of operational checks.

The first check is *band defect*. The retained finite-dimensional object must approximate the full survival or continuation profile on the declared cycle window. The defect $\varepsilon_{\mathrm{band}}$ records the part of the profile left outside the retained band. A small band defect says that the finite-band description is an adequate low-dimensional interface object at tolerance η.

The second check is *scalarity defect*. Even if the retained band is accurate, its principal scalar may not dominate the retained profile. The defect $\varepsilon_{\mathrm{scalar}}$ records the visible contribution of non-principal retained modes under the declared readout class $\mathcal{R}$. A small scalarity defect sends the signature to stratum 1 by the precedence convention; a large scalarity defect forces the typology to ask whether the finite-band object is interpretable stratum 2a, stratum 3, or merely formal-cone non-diagnostic.

The third check is *multiplicativity or composition defect*. When several blocks are composed, the typology only uses composition laws for which the required compatibility data have been supplied. Failure of multiplicativity, failure of shared basin-coordinate semantics, or failure of scalarization to commute with a proposed pooling operation is evidence that the proposed composition does not belong to the currently supplied signature data C.

The fourth check is *norm or normalization defect*. All finite-band, scalarity, and readout estimates are norm-relative. A bound that is harmless in an $L^2(\pi)$ representation may not be harmless after a readout with small principal overlap, and a scalarization meaningful only up to projective normalization cannot be compared as though its absolute scale were canonical.

These four checks interact, but this paper does not assert a universal additive error formula among them. The operational instruction is simpler: keep the defect labels separate, keep the declared tolerance η fixed with the signature, and state which estimates or supplied compatibility data are being used whenever a finite-band object is treated as scalar, comparable, or composable.

7.2 Uniform scalarity defect over a readout class

We now record the basic uniform estimate behind the scalarity-defect slot. The statement is readout-relative: scalarity depends on both the retained semigroup and the class of readouts through which unresolved modes remain visible.

Let G be a reversible killed generator on a finite state space, viewed in an orthonormal eigenbasis $\phi_1, \ldots, \phi_N$ of $L^2(\pi)$, with

$$G\phi_r = -a_r\phi_r, \qquad 0 < a_1 < a_2 \le \cdots \le a_N.$$

Write the survival profile at cycle length Δ as

$$q_\Delta = e^{\Delta G}1 = \sum_{r=1}^{N} e^{-a_r\Delta} b_r\phi_r, \qquad b_r := \langle 1, \phi_r\rangle_\pi.$$

For a retained band $E_K = \operatorname{span}\{\phi_1, \ldots, \phi_K\}$ and a readout $\sigma \in E_K^*$, define the readout-relative scalarity defect

$$D_K(\sigma, \Delta) := \frac{\left|\sum_{r=2}^{K} e^{-a_r\Delta} b_r\sigma(\phi_r)\right|}{e^{-a_1\Delta}\,|b_1\sigma(\phi_1)|},$$

whenever the denominator is nonzero. This is the relative size, under σ, of the visible retained non-principal contribution compared with the principal scalar contribution.

Lemma 7.1 (Uniform scalarity-defect bound)**.** *Let $\mathcal{R} \subset E_K^*$ be an admissible readout class, and choose normalized representatives of the projective class, for example $\|\sigma\|_{E_K^*} = 1$ for all $\sigma \in \mathcal{R}$. The bound below is a statement about this normalized class. Assume a principal-overlap margin*

$$\inf_{\sigma\in\mathcal{R}} |b_1\sigma(\phi_1)| \ge m > 0,$$

and modal-visibility bounds

$$\sup_{\sigma\in\mathcal{R}} |b_r\sigma(\phi_r)| \le M_r \qquad (2 \le r \le K).$$

Then

$$\sup_{\sigma\in\mathcal{R}} D_K(\sigma, \Delta) \le \sum_{r=2}^{K} \frac{M_r}{m}\, e^{-(a_r-a_1)\Delta}.$$

In particular, the scalarity defect is uniformly small over $\mathcal{R}$ whenever the non-principal retained modes are suppressed on the cycle window and the principal-mode readout margin does not degenerate.

Proof. For fixed $\sigma \in \mathcal{R}$,

$$D_K(\sigma, \Delta) \leq \sum_{r=2}^{K} e^{-(a_r - a_1)\Delta} \frac{|b_r \sigma(\phi_r)|}{|b_1 \sigma(\phi_1)|}$$

by the triangle inequality. The denominator is bounded below by m, and each numerator is bounded above by M_r. Taking the supremum over $\sigma \in \mathcal{R}$ gives the displayed estimate. □

The principal-overlap margin is load-bearing. If admissible readouts can become nearly orthogonal to the principal mode, a relative principal-scalar approximation can fail uniformly even when the spectral gap is large. This is why $\varepsilon_{\text{scalar}}$ is recorded together with the readout class $\mathcal{R}$ rather than as a purely spectral number.

7.3 Supplied composition data

The signature component C records composition-compatibility data supplied with the modeling setup. The typology uses these data when supplied; in their absence it does not infer compositions from signature components alone.

Examples of possible C-data include: shared basin coordinates across blocks, a product-basin interpretation for a cluster of witnessed blocks, a declared scalarization to be applied before outer aggregation, a chosen embedding of scalar blocks into a larger cone, or an explicit product-typed interface object. These data can make a proposed composition meaningful. Without them, the typology refuses to infer a composition law from the mere fact that two blocks each have signatures.

This discipline is especially important for finite-band objects. Scalar stratum 1 blocks compose by the monograph's geometric evaluator once its scalar hypotheses are met. Interpretable stratum 2a blocks may compose under shared basin-coordinate semantics, but that shared semantics is additional C-data. Stratum 3 blocks carry cone-order partial comparison, not a canonical scalar aggregate. Mixed-type cross-stratum composition is still less automatic: scalarizing a cone-valued block, embedding a scalar block into a cone, or retaining a product-typed interface are different choices with different consequences. Those choices are deferred to the future-work discussion in Section 9.

Thus C is a slot for supplied compatibility structure. The present paper gives an illustrative same-type construction below; it does not prove a generic positive theory of C.

7.4 Compatible basin-coordinate pooling

Example 7.2 (Compatible basin-coordinate pooling)**.** Suppose blocks $i = 1, \dots, n$ are all in interpretable stratum 2a and have shared basin-coordinate semantics. Concretely, each block has a witnessed basin cone identified with strictly positive coordinates

$$v_i = (v_{i,1}, \dots, v_{i,K}) \in (0, \infty)^K,$$

where coordinate α has the same basin meaning across all blocks. Let $w_i > 0$ with $\sum_i w_i = 1$. The supplied C-data may declare componentwise geometric pooling

$$B_\alpha(v_1, \dots, v_n) := \prod_{i=1}^{n} v_{i,\alpha}^{w_i}, \qquad \alpha = 1, \dots, K.$$

This produces a pooled basin-coordinate vector

$$B(v_1, \dots, v_n) := (B_1, \dots, B_K) \in (0, \infty)^K.$$

The strict-positivity assumption keeps the logarithmic and regrouping formulas unambiguous. If zero basin coordinates are admitted, the same construction can be treated by continuous extension as coordinates tend to zero, but then boundary conventions must be recorded explicitly in C.

The construction is regrouping-compatible at the coordinate level. If the blocks are partitioned into clusters $G_1, \dots, G_m$, with cluster weights $W_j = \sum_{i \in G_j} w_i$ and inner weights w_i/W_j, then pooling inside each cluster and then pooling clusters gives, for each coordinate α,

$$\prod_{j=1}^{m} \left(\prod_{i \in G_j} v_{i,\alpha}^{w_i/W_j} \right)^{W_j} = \prod_{i=1}^{n} v_{i,\alpha}^{w_i}.$$

Thus the componentwise pooling operation has the same formal regrouping property as the scalar weighted geometric mean, but only because the shared basin-coordinate interpretation was supplied. Note that this regrouping is a coordinate-level property; it does not transfer to scalar readouts in general. For a linear basin-mass readout $r(v) = \sum_\alpha m_\alpha v_\alpha$, in particular,

$$r\big(B(v_1, \dots, v_n)\big) \neq \prod_{i=1}^{n} r(v_i)^{w_i}$$

in general, so a scalar reduction after componentwise pooling is not the same as componentwise pooling of scalar reductions. Whether the scalar readout commutes with the pooling operation is a separate piece of supplied C-data.

This example does not assert a generic outer theorem for interpretable stratum 2a. Nor does it prove that a basin-mass readout is multiplicative or homomorphic with respect to componentwise pooling. If a scalar is extracted after pooling, the scalar readout and its compatibility with the pooling operation must also be part of the supplied C-data. The example shows what compatible same-type data can look like; cross-stratum composition is a separate future-work problem.

7.5 Subspace, coordinate, and basin-label continuity

The final operational lemma records a perturbation-theoretic distinction that the typology uses repeatedly. Three distinct stability properties matter for the typology: stability of a retained spectral subspace, stability of coordinates in that subspace, and stability of basin labels. These are increasingly stringent, and the typology distinguishes them.

Lemma 7.3 (Fixed-rank retained-subspace continuity)**.** *Let A_s be a continuous path of linear operators on a finite-dimensional vector space E, for s in an interval I. Suppose that for each s a cluster of K eigenvalues is separated from the rest of the spectrum by a contour Γ_s that may be chosen locally continuously in s, with the retained rank K fixed. Let*

$$P_s = \frac{1}{2\pi i} \int_{\Gamma_s} (zI - A_s)^{-1} \, dz$$

be the Riesz projection onto the retained spectral subspace. Then $s \mapsto P_s$ is continuous locally on I.

If, in addition, the retained eigenvalues do not cross and remain individually isolated, local eigenvectors or coordinate bases can be chosen continuously after gauge choices such as signs, phases, or normalizations. This coordinate continuity is extra structure beyond subspace continuity. Basin-label continuity is still further structure: it requires an identification of the retained coordinates with metastable basins or basin-local profiles, and is not implied by either P_s-continuity or coordinate continuity alone.

Proof. The continuity of P_s is the standard finite-dimensional Riesz-projection perturbation theorem: the resolvent $(zI - A_s)^{-1}$ varies continuously with s on contours separated from the spectrum, and the contour integral therefore varies continuously [10]. In killed Markov-process settings, perturbation results for killed processes and quasi-stationary distributions give model-specific versions of the same retained-projection stability [17]. The assertions about coordinate bases and basin labels are not consequences of the Riesz-projection theorem; they state the additional data required to move from subspace tracking to coordinate tracking and then to basin interpretation. □

The operational slogan is therefore:

$$\text{subspace continuity} \neq \text{coordinate continuity} \neq \text{basin-label continuity.}$$

Section 8 recalls this distinction when discussing moving coarse states and intermediate dynamic regimes. It is also the reason weak-coupling stability of retained projections, by itself, does not settle stability of interpretable stratum 2a.

8 Dynamic regimes

The dynamic axis $J \in \{\mathrm{qs}, \mathrm{fz}, \mathrm{int}\}$ was introduced in Section 2.3 as part of the signature, separating static stratum data from dynamic-regime data by the semicolon in Σ. Section 3 stated that J refines but does not replace the static stratum. This section develops what the three regimes mean operationally, why intermediate regimes can amplify pluralism, and how the perturbation-theoretic continuity distinctions of Section 7 apply to dynamic continuation.

8.1 The J-axis revisited

Let q^* denote the relevant coarse-state transition or mixing rate, as appropriate to the regime, and let κ denote a viability-relevant killing or continuation rate; in spectral models, κ is the local principal killing rate $a_1(z)$ of Section 7. The dynamic regime J records how these two rates relate on the declared cycle window of length Δ.

The label J is orthogonal to the static stratum: a stratum-1 signature can occur in any of the three regimes, as can a stratum-3 signature. The static stratum records what kind of interface object the retained band supplies; J records how that object enters a continuation problem when the coarse state is allowed to evolve over the cycle. The signature thus separates two questions: what is the type of the retained interface, and what dynamic context is the typology asked to evaluate it in.

8.2 Quasi-static and fast-coarse-mixing regimes

In the quasi-static regime qs, the coarse state is effectively frozen over the cycle: $q^*\Delta \ll 1$. Static spectral and cone data can be read directly, with corrections of order $q^*\Delta$ in the relevant readout-relative norms. A stratum-1 signature in qs uses the monograph's geometric evaluator on per-state scalar values; a stratum-3 signature in qs uses the cone-order partial comparison with negligible coarse-state evolution within the cycle. Static strata apply with quantifiable corrections in this regime.

In the fast-coarse-mixing regime fz, the coarse state mixes rapidly compared with the principal killing/continuation scale but still slowly compared with the retained-band compression gap. In the notation of Proposition F.11 of [11], this is the triple separation

$$\max_z a_1(z) \ll q^* \ll g^*,$$

where g^* is the retained-band compression gap. The fast-coarse-mixing window therefore lies strictly inside the slow-fast hypotheses, with q^* bounded above by g^*. Integrated killing rates self-average to effective rates inside this window, and the typology evaluates the static strata against these averaged rates rather than the bare per-state values. Within the window, fz is the regime where averaging-based simplification is robust.

Both qs and fz thus return the dynamic continuation problem to a static or quasi-static evaluation. The static stratum classification carries through with model-dependent corrections controlled by the relevant slow-fast estimates.

8.3 Intermediate regimes and the Jensen gap

The intermediate regime int is the case in which $q^* \sim \kappa$ on the cycle window. Neither the quasi-static reduction nor the fast-mixing average is available before conditioning on the coarse-state trajectory. The continuation problem becomes genuinely path-dependent: viability-relevant survival is given by a Feynman–Kac expectation

$$\mathbb{E}\left[\exp\left(-\int_0^\Delta \kappa(X_s)\,ds\right)\right],$$

where X_s is the coarse-state trajectory and the expectation is over its law on the cycle window.

The Jensen gap is generally nontrivial. Convexity of $\exp(-\cdot)$ gives

$$\mathbb{E}\left[\exp\left(-\int_0^\Delta \kappa(X_s)\,ds\right)\right] \;\geq\; \exp\left(-\mathbb{E}\left[\int_0^\Delta \kappa(X_s)\,ds\right]\right),$$

with equality only when the path integral $\int_0^\Delta \kappa(X_s)\,ds$ is almost surely constant. Whenever the coarse-state evolution and the killing-rate variation produce path variance, the survival probability strictly exceeds the average-rate proxy. Static-stratum analysis applied to the average rate alone misses this gap, and the typology cannot recover the int-regime continuation from per-state scalarizations alone.

This is where the matrix-valued Feynman–Kac structure of the program becomes load-bearing. In finite-band settings the scalar potential $\kappa(X_s)$ is replaced, in the deferred theory, by retained-band transport and mode-dependent continuation data; this is the matrix-valued extension flagged in Section 9 as future work F2. The typology records the $J = \text{int}$ label here without developing the technical machinery.

8.4 Dynamic regimes can amplify pluralism

The claims in this subsection are typing consequences and warnings; the matrix-valued continuation theorem itself is deferred to F2 in Section 9.

The intermediate regime carries a structural consequence for the static typology. Even when each frozen coarse state supplies an operationally scalar local interface in the stratum-1 sense, the intermediate-regime continuation need not reduce to a scalar obtained by averaging those local interfaces before continuation; the scalar, when it exists, is the post-continuation Feynman–Kac value $\mathbb{E}[\exp(-\int_0^\Delta \kappa(X_s)\,ds)]$, and the Jensen gap above is the gap between this scalar and the average-rate proxy.

For stratum-3 signatures the amplification is stronger. Cone-order partial comparison at each instant must be tracked through path-dependent continuation; the retained cone or its representation may vary along the trajectory, and admissible scalarizations $\sigma \in \mathcal{R}$ that respect the cone at one time may not respect it at another. The forced-pluralism content of stratum 3 is preserved, amplified, or reconfigured depending on how the cone and admissible class transport along the dynamics.

For interpretable stratum 2a the question is more delicate. The witness data of Criterion 5.1 are stated for a fixed signature on a declared cycle window; whether basin geometry, basin-local profiles, basin-mass readout, and cycle-window stability survive intermediate dynamic regimes is an open problem. Section 9 flags this as future work F3.

8.5 Subspace, coordinate, and basin-label continuity in the dynamic axis

Lemma 7.3 of Section 7 distinguished three increasingly stringent stability properties: subspace continuity, coordinate continuity, and basin-label continuity. The same distinction governs how a signature behaves under variation of the dynamic regime label.

When the coarse-state dynamics vary continuously — in a perturbation of the generator, a slow change in the cycle window, or a smooth transition between qs, int, and fz — the retained spectral subspace can vary continuously under fixed-rank and non-crossing hypotheses. Coordinate continuity requires further gauge choices. Basin-label continuity requires the additional identification with metastable basins, and is not implied by either of the previous two.

For dynamic regime transport this means: weak-coupling stability of retained projections or retained subspaces, by itself, does not settle whether interpretable stratum 2a survives a continuous deformation of the signature's dynamic context. The signature's J label may transition smoothly while the basin-witness data are reorganized or lost. This is the technical core of the open question on dynamic stability of interpretable 2a, deferred to Section 9.

9 Future work and program implications

The typology developed in Sections 2–8 resolves the central question raised in the introduction: a readout-insensitive finite-band middle (the dissolved 2b proposal) cannot be sustained on rich comparison domains, and the surviving regions are interpretable stratum 2a, stratum 3, the formal-cone non-diagnostic region, and stratum 0, refined by the dynamic axis $J \in \{\mathrm{qs}, \mathrm{fz}, \mathrm{int}\}$. This section flags what the typology does not resolve. The first part names four technical arcs deferred to subsequent program work. The second part states the consequences for the surrounding documents in the program: the monograph (Document 1) and the empirical extraction methodology (Document 3).

9.1 Technical deferrals

Four arcs are flagged as future technical work. Each is recorded as a deferral with its hypotheses sketched; none is developed here.

F1. Weak-coupling cone and basin stability. Lemma 7.3 gives fixed-rank retained-subspace continuity under non-crossing eigenvalue clusters. The arc on which this lemma extends to the cone $\mathcal{K}$ and the basin-witness data (W_1)–(W_5) is open. The natural questions: when a stratum-3 signature is weakly coupled to neighboring blocks, do the cone $\mathcal{K}$ and dual cone $\mathcal{K}^*$ remain stable in an appropriate metric? When a witnessed interpretable-2a block is weakly coupled, do the basin family, basin profiles, basin-mass readout, and cycle-window stability survive the perturbation? Polyhedral cones with explicit angular margin and basin geometries with explicit return-time bounds are the tractable special cases. F1 collects these questions without resolving them.

F2. Matrix-valued Feynman–Kac for the retained band. Section 8.3 stated the scalar Feynman–Kac form for intermediate-regime continuation. The matrix-valued extension — replacing the scalar potential $\kappa(X_s)$ with retained-band transport and mode-dependent continuation data — is F2. The natural setup, schematically, is a coarse-state-dependent operator on the retained band, with the continuation problem reading as an ordered-exponential or path-ordered product expectation. The retained band becomes a vector bundle over the coarse-state space, requiring fixed-rank, non-crossing, and gauge/connection data for continuation along trajectories. F2 is the dynamic-axis analog of the static finite-band typology. Its development is deferred to subsequent program work.

F3. Dynamic stability of interpretable 2a. Section 8.5 flagged the open question whether basin-witness data of Criterion 5.1 survive intermediate dynamic regimes. F3 is the dedicated arc. Two sub-arcs are anticipated: a positive direction giving sufficient conditions for basin-canonical scalarization to persist along continuous deformations (likely requiring matrix-valued continuation data of F2), and an obstruction direction characterizing when no basin-canonical scalarization survives even though the retained subspace remains continuous. Negative results are also possible: basin labels may fail to be preserved under dynamics that preserve subspaces and even coordinates, recovering the three-tier slogan of Lemma 7.3 in a dynamic setting.

F4. Cross-stratum composition. Example 7.2 supplied a same-type composition construction under explicit shared-coordinate hypotheses. Cross-stratum composition is more delicate: the inputs to a composition may belong to different strata, and combining them requires type-conversion data. Four resolutions are recorded without commitment, listed in roughly increasing typological cost:

(i) scalarize a higher-type interface (interpretable 2a or stratum 3) into stratum 1 by a supplied scalar readout, accepting the loss of partial-order content;

(ii) embed a stratum-1 scalar into a shared cone-valued representation in stratum 3, accepting that the embedding is additional supplied structure not derivable from the scalar block;

(iii) retain a product-typed interface $(c, v) \in \mathbb{R}_+ \times \mathcal{K}$ across mixed-stratum blocks, treating the product structure as the composed signature;

(iv) prove a generic promotion to stratum 3 under appropriate hypotheses — the only theorem-shaped option, and the most ambitious.

None of these is generic; each requires supplied compatibility data and has different consequences for the typology. F4 is the arc on which type-conversion options apply when, and what they cost.

9.2 Program implications

The typology has consequences for the surrounding documents in the program.

Document 1: rank-one scalar realization. The monograph develops the rank-one scalar realization of the program in detail: the geometric evaluator, the scalar Feynman–Kac synthesis, viability-readout discipline, and Axiom A5 as the structural condition for scalar interface sufficiency. Document 2 surrounds this content. The typology identifies stratum 1 as the regime in which the monograph's scalar machinery applies, and locates the surrounding regions (interpretable 2a, stratum 3, formal-cone non-diagnostic, stratum 0) as the cases where rank-one scalar compression is not adequate. Theorem 4.3 of Section 4 identifies the retained-band structural prerequisite for evaluator-A5, and thereby marks the boundary of stratum 1 inside finite-band settings. The monograph's outer aggregation theorem applies to scalar outputs in stratum 1, and applies coordinatewise to compatible same-type stratum-2a constructions (Example 7.2) only after C-data declare each basin coordinate to be a positive scalar interface channel; scalar readouts after pooling require additional supplied compatibility data. Cross-stratum composition is F4 territory.

The framing is conservative. The monograph's mathematics is unchanged. Document 2's contribution is to record where that mathematics applies, what surrounds it, and what data must be supplied for the surrounding regions.

Document 3: empirical signature extraction. Document 2 supplies signature data $\Sigma = (K, a, \varepsilon_{\text{scalar}}, \varepsilon_{\text{band}}, \mathcal{R}, C, \eta; J)$ but does not extract it from observation. Document 3 develops the empirical methodology for that extraction. Four classes of methods are anticipated:

(1) *Sublethal perturbation tomography* — controlled sublethal perturbations across components or modules, with response measured against viability-relevant quantities, used to identify retained-band dimension K, defect terms $\varepsilon_{\text{band}}, \varepsilon_{\text{scalar}}$, and the spectral or structural data behind them.

(2) *Basin reconstruction* — identification of metastable basin structure from sampling, time-series, or perturbation response, with the recovered basin geometry providing candidate witness data $(W_1), (W_2)$ for Criterion 5.1.

(3) *Hysteresis and multistability assays* — experimental probes for whether the system supports multiple basins or alternative steady states, providing evidence for candidate $K \geq 2$ signatures and tests for cycle-window stability (W_5).

(4) *Cone-comparability versus scalarization-dependent comparison tests* — tests of whether observed orderings are unanimous across a declared admissible class $\mathcal{R}$ (cone-comparable) or scalarization-dependent (stratum-3 territory with proper $\mathcal{R} \subsetneq \mathcal{K}^*$).

Document 3 does not commit to specific biological systems. The extraction methodology is general and applies across the motivating cases discussed in the program-level documents (cell-, organism-, colony-, ecosystem-scale signatures) without adjudicating which ones instantiate which strata. That adjudication is the empirical work itself.

Section 10 closes by recapping the narrower-stronger thesis and the open frontier the program leaves for Document 3 and the technical arcs above.

10 Closing

Document 2 has done two things. It typed the interface between local viability and outer aggregation as a signature, and it dissolved a candidate finite-band stratum that would have offered a readout-insensitive scalar middle. The dissolution is the paper's central technical contribution: Theorem 4.1 closes the readout-insensitivity escape route for linear readouts on rich comparison domains, and Theorem 4.3 closes the temporal-sufficiency escape route via the retained-band structural prerequisite for evaluator-A5. The surviving regions are five — stratum 0, stratum 1, interpretable stratum 2a, the formal-cone non-diagnostic region, and stratum 3 — refined by the dynamic axis $J \in \{\text{qs}, \text{fz}, \text{int}\}$.

This narrows the program's scalar reach and sharpens its formal commitments. The narrowing: rank-one scalar compression is one regime among five, and the dissolved 2b conjecture is unavailable on rich domains. The sharpening: the surviving regions carry substantive partial-order content recorded by signature data. Stratum 3 in particular is forced pluralism with mathematical content. Cone-comparable pairs are unanimously rankable across the full dual cone $\mathcal{K}^*$, and hence across any declared admissible subfamily $\mathcal{R} \subseteq \mathcal{K}^*$; cone-incomparable pairs reflect what the dynamics themselves do not totally order. A restricted $\mathcal{R}$ may impose additional unanimous-$\mathcal{R}$ comparisons on cone-incomparable pairs; those rankings are observer-indexed through the declared readout class, distinct from cone-order comparability. Forced pluralism is the result the typology delivers, with the cone $\mathcal{K}$ recording structural partiality and the admissible class $\mathcal{R}$ recording the observer- or measurement-indexed restriction.

The monograph (Document 1) is located by Document 2 as the rank-one scalar realization, with mathematics unchanged. Document 3 will take the typology to data, developing extraction methodology for signature components from observation. F1–F4 collect the technical arcs that subsequent program work must address: weak-coupling cone and basin stability, matrix-valued Feynman–Kac, dynamic stability of interpretable 2a, and cross-stratum composition. The typology presented here is the type discipline through which the program can now pose its empirical and dynamical questions.

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